

Sri Lanka Institute of Information Technology



AutoFlow Sri Lanka: Vehicle Service & Fuel Management System

Project Proposal Report

Module Code: SE2030

Module Name: Software Engineering

Academic Year: 2026

Semester: Year 2 – Semester 1

Group ID: 2026-Y2-S1-KU-01

Name	Student ID	Assigned Major Function
Manathunge A.I	IT23642096	Vehicle Management
Ahamed M.N.A	IT24104390	Service Booking Management
Sarma S. S. L.	IT25100002	Fuel Management
Nayanajith K.A.T	IT25100020	Payment Management
Herath H.M.A.R.B	IT25100117	Notification Management
Liyanage D.D.S	IT25100119	User Management

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Requirement Gathering Process and Evidence

The team conducted requirement gathering through the SE2030 Project Requirement Discussion Bot. Stakeholders were consulted to understand the problems with manual vehicle-service, fuel-recording, booking, payment, and communication processes. The gathered responses were used to identify users, workflows, functional requirements, and system constraints.

SAMPLE 01

Dilmi / Customer Service · booking, cancellation, refund, complaint and customer data

thank you for your time. this is Dilum. Dilmi I want you to go through the tasks that you do daily, and how you do it.

 Project Discussion Bot - Year2
Dilmi Jayasinghe – Customer Service Executive
Hello Dilum. Thank you for taking the time to speak with me.
My day usually starts by checking new customer inquiries, service requests, and notifications received overnight. I log into the system and review any pending tasks that need attention.
When a customer wants to register a vehicle, I collect details such as the vehicle number, owner's name, contact details, vehicle type, and current mileage. I enter these details into the system and verify that there are no duplicate records.
A large part of my day is handling service bc ↓ jgs. Customers call, send messages, or visit our office to book a service. I check available dates, create the booking, and provide confirmation to

explain how you do each thing that you mentioned right now?
 Project Discussion Bot - Year2
Dilmi Jayasinghe – Customer Service Executive
Hello Dilum. Sure, I will explain each task step by step as I do it in my daily work.
Registering a New Vehicle
When a customer comes to our office or calls us, I first collect their details. I ask for the owner's name, phone number, address, vehicle registration number, vehicle type, make and model. I also ask for the current mileage.
Then I search the system to check whether the vehicle already exists. If it does not exist, I create a new vehicle record and save all the information. Finally, I verify the details with the customer before saving.
Updating Vehicle Details ↓

SAMPLE 02

Niroshan / Operations · access, authority, breakdowns and business rules

Niroshan, This is Anjana. Please let me know how you do all the tasks now?

 Project Discussion Bot - Year2
Niroshan Perera – Workshop Operations Manager
Hello Anjana. Sure, I will explain exactly how I do my daily tasks.
When I start work, I open the system dashboard and check all the service bookings scheduled for the day. I look at the vehicle details, service type, booking time, and workshop bay availability. This helps me understand the workload for the day.
When a vehicle arrives, I search for the booking in the system and confirm the vehicle registration number and customer details. If everything is correct, I mark the vehicle as arrived and assign it to an available service bay. ↓

consider all the facts that we have discussed so far and suggest anything that we may have missed.
 Project Discussion Bot - Year2
Niroshan Perera – Workshop Operations Manager
Hello Anjana. After our discussion, I think you have covered the main functions well. However, there are a few things that may have been missed during requirement gathering.
For Vehicle Management, we should keep vehicle documents such as insurance, revenue licence, and registration expiry dates. The system should remind users before these documents expire.
For Service Booking Management, it would be useful to record the service type, service priority (normal or urgent), technician assigned, and ↓ mated completion date.

SAMPLE 03

Ruwan / Scheduling · bay/technician allocation, demand and change impact

ruwan im Anjali, do you also handle customer data for scheduling decisions?

 Project Discussion Bot - Year2
Ruwan Fernando – Service Scheduling Officer
Hello Anjali. Yes, I do use some customer data when making scheduling decisions, but only the information needed for booking and service operations.
When I schedule a service, I check the customer's vehicle details, previous service history, preferred appointment times, and any special requests they have made. This helps me assign the correct service type and estimate how much workshop time is needed.
For example, if a customer's vehicle had a mainr repair recently, I may allocate a longer inspection slot. If they regularly request mor. ↓ j appointments, I try to accommodate that if share is available

ruwan this is Anjali. explain what are the difficulties you are facing
 Project Discussion Bot - Year2
Ruwan Fernando – Service Scheduling Officer
Hello Anjali. Yes, there are several difficulties I face in my daily work.
One common problem is **last-minute cancellations and rescheduling requests**. Sometimes customers cancel on the same day, which leaves workshop time slots unused. On the other hand, some customers want urgent appointments when the schedule is already full.
Another challenge is **limited workshop bay availability**. There are times when many vehicles need servicing at once, but we do not have enough bays or technicians available. This makes scheduling more difficult.
I also face issues when **vehicle information or customer details are incomplete or outdated**. If ↓

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SAMPLE 04

Malsha / Finance · revenue, payment, reconciliation, refund and approval

thank you for your time. this is Lojan. Malsha I want you to go through the tasks that you do daily, and how you do it.

Project Discussion Bot - Year2

Malsha Wickramaratne – Finance & Revenue Manager

Hello Lojan. Thank you for speaking with me.

My main responsibility is managing payments, revenue records, and financial tracking. Every day, I use the system to monitor customer payments and ensure all financial transactions are recorded correctly.

When a customer makes a payment, I first search for the related invoice. I verify the amount due and the payment method used, such as cash, card, or bank transfer. Then I confirm the payment and update its status in the system.



thank you for your time. this is Lojan. Malsha I want you to go through the tasks that you do daily, and how you do it.

Project Discussion Bot - Year2

Malsha Wickramaratne – Finance & Revenue Manager

Hello Lojan. Thank you for speaking with me.

My main responsibility is managing payments, revenue records, and financial tracking. Every day, I use the system to monitor customer payments and ensure all financial transactions are recorded correctly.

When a customer makes a payment, I first search for the related invoice. I verify the amount due and the payment method used, such as cash, card, or bank transfer. Then I confirm the payment and update its status in the system.



I also check for failed or pending payments. If a payment attempt has failed, I review the reason

SAMPLE 05

Kasun / Fuel Station · daily work, dispensing, incidents and live dashboard

Kasun, I am Amantha. please go through everything that you do daily and how you do them

Project Discussion Bot - Year2

Kasun Senanayake – Fuel Station Manager (Colombo Depot)

Hello Amantha. I'd be happy to explain my daily work and how I do each task.

My day starts by checking the fuel inventory levels in the system. I review how much petrol and diesel are currently available in the depot and compare them with the previous day's closing balance. If fuel stock is getting low, I inform management so they can arrange a new fuel delivery.

When a company vehicle comes for refuelling, I first identify the vehicle using its registration number. I check the vehicle record in the system and verify that it is authorised to receive fuel.

After the fuel is pumped, I create a fuel-log . I enter the vehicle number, fuel type, quantity issued, current odometer reading, date, time, and fuel cost. Once saved, the system

Kasun here again, are you preparing any reports daily. if so can you provide me an example

Project Discussion Bot - Year2

Kasun Senanayake – Fuel Station Manager (Colombo Depot)

Hello Amantha. Yes, I prepare a **Daily Fuel Usage Report** at the end of every day. This report helps me monitor fuel issues, vehicle consumption, and remaining stock.

Here is a simple example:

Daily Fuel Usage Report

Date: 24/07/2026



Vehicle No	Fuel Type	Quantity (L)	Odometer (km)	Cost (LKR)	Driver
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SAMPLE 06

Niroshan / Validation · roles, security, audit, capacity and retention

hey Niroshan, Im Afsal. I will be attending to gather requirements for the system you want, please explain what needs to be done. what are the day to day tasks you do?

Project Discussion Bot - Year2

Niroshan Perera – Workshop Operations Manager

Hello Afsal. Thank you for gathering requirements for us. My main responsibility is making sure all vehicle services are completed on time and that the workshop operates smoothly.

A typical day starts with me checking all service appointments scheduled for the day. I review the vehicles coming in, the type of service required, available technicians, and available service bays. Based on the workload, I plan how work should be assigned.

When a vehicle arrives, I verify the booking  and assign the vehicle to a suitable service bay. Then I assign a technician who has the right skills for that service.

do you currently have a system for this? if so let me know I need to observe

Project Discussion Bot - Year2

Niroshan Perera – Workshop Operations Manager

Hello Afsal.

Yes, we currently use a basic system, but it is quite limited. It allows us to record vehicle service bookings, update service status, and maintain some vehicle service history. However, many activities still require phone calls, spreadsheets, or manual coordination between departments.

If you were observing my work, I would show you:

1. **Daily Service Schedule Screen**
 - List of bookings for the day



1. Introduction

1.1 Project Overview

AutoFlow Sri Lanka is a web-based Vehicle Service and Fuel Management System designed to connect vehicle owners, service centres, fuel stations, and administrative staff through one integrated platform. The system will replace manual and disconnected processes such as paper job cards, phone-call bookings, handwritten fuel logs, separate invoices, and spreadsheet-based reports.

Through AutoFlow Sri Lanka, customers can register their vehicles, book service appointments, record fuel-related information, make payments, receive digital invoices, and receive important notifications. Staff members can manage bookings, vehicle records, payments, fuel records, customer accounts, and operational information according to their assigned roles.

1.2 Problem Statement

Many vehicle service centres and fuel stations still depend on manual records and separate communication methods. This causes duplicate data entry, booking conflicts, delays in customer updates, difficulty in tracking fuel usage, and unreliable financial reporting. Customers may also face inconvenience when they need to call or visit a service centre to make, cancel, or reschedule an appointment.

Therefore, a centralized system is needed to improve coordination, reduce errors, and provide accurate information to both customers and staff.

1.3 Objectives

The main objectives of AutoFlow Sri Lanka are:

- To provide a centralized platform for vehicle, service booking, fuel, payment, notification, and user management.
- To allow customers to make service bookings online and manage their appointments easily.
- To reduce scheduling conflicts by checking service-bay and technician availability.
- To maintain accurate vehicle and fuel records for better monitoring.
- To support verified payments, digital invoices, refunds, and transaction tracking.
- To provide timely notifications about bookings, payments, delays, cancellations, and service reminders.
- To improve security through role-based access control and audit logs.

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1.4 Target Users

The main users of the system are registered customers, guest customers, customer-service officers, scheduling officers, finance officers, fuel-station managers, workshop operations managers, administrators, and senior management. External stakeholders such as banks, payment gateways, fuel suppliers, insurance providers, regulators, and IT support staff may also interact with or support the system.

1.5 Scope and Limitations

The first version of the system will focus on six core modules: Vehicle Management, Service Booking Management, Fuel Management, Payment Management, Notification Management, and User Management. It will be developed as a web-based system that can be accessed through modern desktop and mobile browsers.

The system will depend on internet access and third-party services such as payment gateways and SMS/email providers. Advanced features such as a mobile application, loyalty programmes, advanced analytics, and group bookings are outside the scope of the initial version.

2. System Overview Diagram

The AutoFlow Sri Lanka system is a centralized web-based platform that connects customers, service-centre staff, fuel-station staff, finance staff, and administrators. Users access the system according to their roles and permissions.

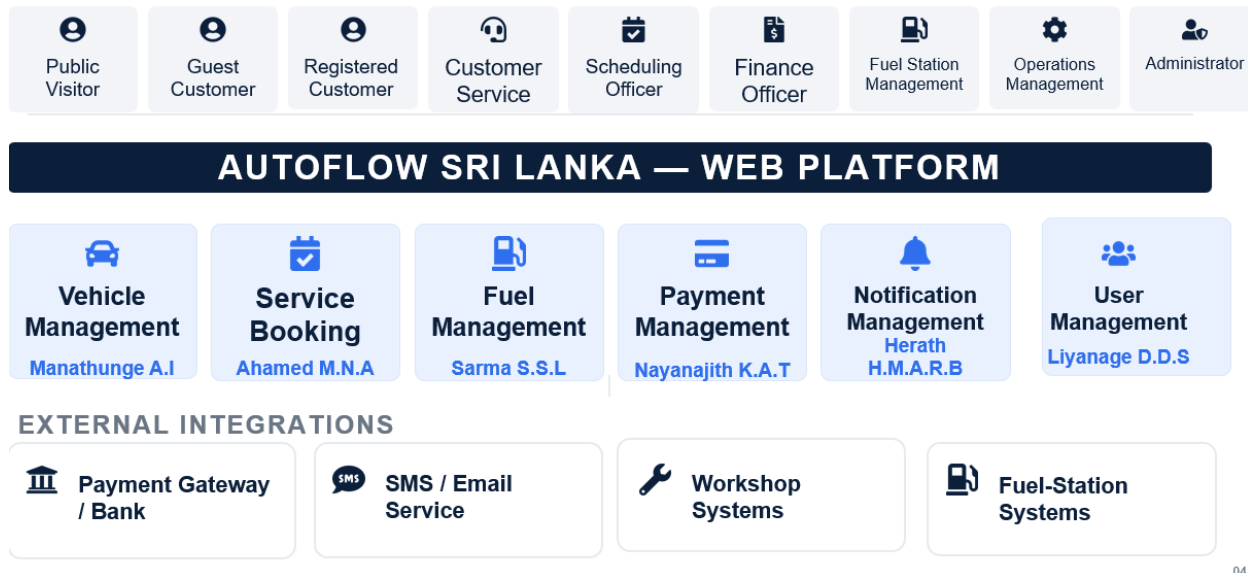
Customers can register and manage their vehicles, book service appointments, make payments, view invoices, review service history, and receive notifications. Service and scheduling officers manage appointment slots, bays, technicians, queues, cancellations, and rescheduling. Fuel-station managers record fuel transactions and monitor fuel consumption. Finance officers verify payments, issue invoices, reconcile transactions, and process refunds. Administrators manage user accounts, roles, and system access.

The platform integrates with external services such as payment gateways or banks, SMS/email services, workshop systems, and fuel-station systems. This integration enables secure payments, automated customer notifications, and accurate operational records.

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ARCHITECTURE

System Overview



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Figure 1: High-Level System Overview of AutoFlow Sri Lanka.

3. Functional Requirements

The AutoFlow Sri Lanka system shall provide the following core functional requirements:

ID	Functional Requirement	Description
FR-01	User Registration and Login	The system shall allow customers and staff to register, log in, log out, reset passwords, and manage their profiles.
FR-02	Role-Based Access Control	The system shall provide different access permissions for customers, service staff, scheduling officers, finance officers, fuel-station managers, administrators, and management.
FR-03	Vehicle Management	The system shall allow users to register, view, update, and manage vehicle details, ownership information, insurance status, mileage, and service history.
FR-04	Service Booking Management	The system shall allow customers to select a vehicle, service type, service centre, date, and available time slot to make a booking.
FR-05	Booking Updates	The system shall allow authorized users to confirm, reschedule, cancel, and manage service bookings and waiting queues.
FR-06	Availability Management	The system shall check the availability of service bays and technicians to prevent booking conflicts.
FR-07	Fuel Management	The system shall record fuel quantity, cost, date, odometer reading, and fuel-consumption history for vehicles or stations.

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FR-08	Fuel Discrepancy Detection	The system shall identify abnormal fuel usage, duplicate fuel entries, or invalid odometer readings for review.
FR-09	Payment Management	The system shall create and verify payment transactions, issue receipts and digital invoices, and maintain payment records.
FR-10	Refund and Reconciliation	The system shall allow finance officers to process approved refunds and reconcile payment transactions.
FR-11	Notification Management	The system shall send booking confirmations, service reminders, payment confirmations, cancellation notices, delay alerts, and refund notifications through SMS, email, or in-system messages.
FR-12	Reports and Audit Logs	The system shall provide reports, dashboards, search/filter functions, and audit logs of important user activities and record changes.

Important business rules:

- One service bay and time slot can have only one active booking.
- A payment must be verified before a digital invoice is issued.
- Vehicles with expired insurance or required inspection status cannot make a service booking.
- A fuel entry with an odometer reading lower than the previous reading must be flagged.
- Duplicate payment transaction IDs must be flagged for finance review.
- Critical changes, including booking cancellation and refunds, must generate customer notifications.

4. Non-Functional Requirements

The AutoFlow Sri Lanka system must meet the following quality requirements:

Category	Requirement
Usability	The system shall provide a simple, clear, and mobile-friendly interface that can be used by customers and staff with minimal training.
Accessibility	The system shall support English, Sinhala, and Tamil to serve a wider range of users in Sri Lanka.
Performance	Vehicle searches and record retrieval should be completed within 3 seconds under normal operating conditions. A service booking should be completed within 10 seconds after successful payment verification.
Security	The system shall use secure authentication, password protection, role-based access control, and protected handling of customer and payment information.
Data Privacy	Customers shall only be able to access their own vehicle, booking, payment, and fuel records. Staff access shall be limited according to their job role.
Reliability	The system shall maintain accurate records and prevent duplicate active bookings, duplicate payment transactions, and invalid fuel entries.

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Availability	The system should aim for 99.5% availability, excluding planned maintenance periods.
Backup and Recovery	Important data shall be backed up regularly, and a recovery process shall be available in case of data loss or system failure.
Scalability	The system should be able to support an increasing number of users, vehicles, bookings, and transactions without major performance issues.
Auditability	The system shall keep audit logs for significant activities, including the user, action, date and time, previous value, and updated value where applicable.
Maintainability	The system shall be developed in separate modules so that functions can be maintained, tested, and improved more easily.

5. Major Stakeholders and Users

The AutoFlow Sri Lanka system serves several internal users and external stakeholders. Each role has different responsibilities and access permissions.

Stakeholder / User	Role in the System
Public Visitor / Guest Customer	Can view basic service information and begin registration. Guest access is limited and does not provide access to personal records.
Registered Customer	Registers and manages vehicles, makes service bookings, checks booking status, makes payments, views invoices, reviews service history, and receives notifications.
Customer Service Officer	Assists customers with account-related questions, vehicle records, booking enquiries, and general service support.
Scheduling Officer	Manages service appointments, time slots, service bays, technician availability, waiting lists, cancellations, and rescheduling.
Finance Officer / Finance Manager	Verifies payments, issues invoices and receipts, reconciles transactions, manages refunds, and reviews financial records.
Fuel Station Manager	Records fuel details, monitors fuel usage and consumption, investigates abnormal fuel records, and manages fuel-related disputes.
Workshop Operations Manager	Monitors service-centre operations, vehicle-service progress, workloads, service capacity, and operational reports.
System Administrator	Creates and manages user accounts, assigns roles and permissions, manages account status, and maintains system access control.
Senior Management	Uses dashboards and reports to monitor operational performance, financial information, bookings, and service trends.
Banks / Payment Gateways	External service providers that support online payment processing and payment verification.
SMS / Email Service Provider	External service provider used to deliver booking, payment, reminder, cancellation, and other automated notifications.
Fuel Suppliers, Insurance Providers, and Regulators	External stakeholders that may provide or require information related to fuel records, vehicle compliance, insurance, and regulations.

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IT Support Team	Maintains the system infrastructure, backups, security updates, and technical support.
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Access-control principle: Each user must only access the information and functions needed for their role. For example, customers can access only their own records, while finance staff can access payment-related records but should not change vehicle-service information unnecessarily.

6. Six Major Functions

The system is divided into six major functions. Each group member is responsible for one core module.

Major Function	Assigned Member	Main User(s)	Expected Outcome
Vehicle Management	Manathunge A. I.	Registered Customer, Customer Service Officer, Administrator	Accurate and reliable vehicle records
Service Booking Management	Ahamed M. N. A.	Registered Customer, Scheduling Officer, Workshop Operations Manager	Conflict-free and efficient service scheduling
Fuel Management	Sarma S. S. L.	Fuel Station Manager, Registered Customer, Finance Officer	Reliable and auditable fuel records
Payment Management	Nayanajith K. A. T.	Registered Customer, Finance Officer	Accurate and transparent payment processing
Notification Management	Herath H. M. A. R. B.	Registered Customer, Staff Users, Administrator	Timely communication and status updates
User Management	Liyanage D. D. S.	Customer, Staff User, Administrator	Secure and well-managed system access

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6.1 Vehicle Management — Manathunge A. I.

Users: Registered Customers, Customer Service Officers, and Administrators.

This function allows users to create, view, update, and manage vehicle information. Vehicle details include registration number, make, model, year, ownership details, insurance status, mileage, and service history.

Sub-functions:

- Register a new vehicle.
- View and update vehicle details.
- Link a vehicle to its owner.
- Record mileage and service-history information.
- Check insurance and vehicle inspection/revision status.
- Archive inactive or transferred vehicles.
- Block booking attempts for vehicles with expired insurance or invalid required documentation.

Expected outcome: The system maintains complete, accurate, and trusted vehicle records for customers and staff.

6.2 Service Booking Management — Ahamed M. N. A.

Users: Registered Customers, Scheduling Officers, and Workshop Operations Managers.

This function allows customers to book vehicle services through the system. Customers can choose their vehicle, required service, preferred service centre, date, and available time slot. Staff can manage schedules, service bays, technicians, queues, cancellations, and rescheduling.

Sub-functions:

- Select a vehicle and service type.
- Choose a service centre, date, and time slot.
- Check available bays and technicians.
- Confirm, cancel, or reschedule a booking.
- Manage waiting lists and booking queues.
- Prevent double booking of the same bay and time slot.
- View booking status and service schedules.

Expected outcome: The system provides efficient, organized, and conflict-free service scheduling.

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6.3 Fuel Management — Sarma S. S. L.

Users: Fuel Station Managers, Registered Customers, and Finance Officers.

This function manages fuel-related records. It records fuel quantity, cost, date, odometer reading, and other fuel-usage details. The system also helps identify unusual consumption patterns, duplicate records, and invalid odometer readings.

Sub-functions:

- Record fuel quantity, price, cost, and date.
- Record vehicle odometer readings.
- View fuel consumption history.
- Compare fuel usage across different periods.
- Detect abnormal fuel consumption.
- Flag duplicate entries and lower odometer readings.
- Manage fuel-related disputes or corrections.

Expected outcome: The system provides reliable, traceable, and auditable fuel records.

6.4 Payment Management — Nayanajith K. A. T.

Users: Registered Customers and Finance Officers.

This function allows customers to make payments for vehicle services and enables finance staff to manage related financial records. Payments are verified before invoices are issued, ensuring that the financial information is accurate.

Sub-functions:

- Create payment requests and payment attempts.
- Process payments through an approved payment gateway or bank.
- Verify payment status.
- Generate digital receipts and invoices.
- Maintain payment history.
- Reconcile payment transactions.
- Process approved refunds.
- Flag duplicate transaction IDs for finance review.

Expected outcome: The system ensures secure, accurate, and transparent payment processing.

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6.5 Notification Management — Herath H. M. A. R. B.

Users: Registered Customers, Staff Users, and Administrators.

This function sends automated messages to users about important system events. Notifications may be delivered through SMS, email, or in-system alerts depending on the notification type and user preference.

Sub-functions:

- Send service-booking confirmations.
- Send appointment reminders.
- Send payment confirmations and digital-invoice alerts.
- Send booking cancellation and rescheduling notifications.
- Send service-delay and emergency alerts.
- Send refund-status notifications.
- Allow users to view in-system notifications.

Expected outcome: Customers and staff receive timely, clear, and relevant information, improving communication and service experience.

6.6 User Management — Liyanage D. D. S.

Users: Registered Customers, Staff Users, and System Administrators.

This function manages user accounts, profiles, authentication, roles, and access permissions. It ensures that every user can access only the functions and information that are appropriate for their role.

Sub-functions:

- Register customer and staff user accounts.
- Log in and log out securely.
- Reset passwords and recover accounts.
- Verify users through OTP or email verification.
- Update user profiles.
- Assign and manage user roles.
- Activate, deactivate, or suspend accounts.
- Control user access based on permissions.

Expected outcome: The system provides secure, controlled, and well-managed access for all users.

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7. Minor Functions

In addition to the six major modules, AutoFlow Sri Lanka will include the following supporting functions:

Minor Function	Description
Login and Logout	Allows registered customers and staff users to securely access and exit the system.
Password Reset	Allows users to reset a forgotten password through a secure verification process.
OTP / Email Verification	Confirms a user's identity during registration, password recovery, or sensitive account actions.
Profile Management	Allows users to update basic profile details, contact information, and communication preferences.
Search and Filter	Enables users and staff to quickly find vehicles, bookings, payments, fuel records, and users.
Dashboards	Provides role-based summaries such as upcoming bookings, payment status, fuel usage, notifications, and operational statistics.
Reports	Generates reports related to bookings, payments, service history, fuel consumption, and system activities.
Digital Invoices and Receipts	Allows customers to view, download, or receive invoices and payment receipts electronically.
Feedback and Complaints	Allows customers to submit feedback or complaints about a service, booking, payment, or fuel record.
Maintenance History	Displays previous vehicle-service records to support future service decisions.
Waiting List	Allows customers to join a waiting list when their preferred service slot is unavailable.
Audit Logs	Records important actions, including the user, activity, date and time, and changed information.
Error and Validation Messages	Displays clear messages when users enter invalid data, use an unavailable time slot, or perform an unauthorized action.

These minor functions support the main modules by improving security, usability, communication, record tracking, and day-to-day system operations.

8. System Limitations and Constraints

Although AutoFlow Sri Lanka is designed to improve vehicle-service and fuel-management activities, the first version will operate within the following limitations and constraints:

Area	Limitation / Constraint
Platform	The initial system will be a web-based application. A dedicated Android or iOS mobile application is not included in Version 1.
Internet Dependency	Users require a stable internet connection to access the system, make bookings, receive updates, and complete online payments.
External Integrations	Payment verification, SMS notifications, and email notifications depend on selected third-party payment-gateway, bank, SMS, and email service providers.
Offline Access	The first version will not fully support offline operation or automatic synchronization of offline records.
Payment Rules	Exact refund percentages, approval limits, late-payment rules, and payment-gateway selection must be confirmed with the organization before implementation.
Fuel Supplier Integration	Direct integration with fuel suppliers or their systems may not be available in the initial release. Fuel details may need to be entered manually by authorized staff.
Device and Browser Support	The system will be designed for current web browsers on desktop and mobile devices. Very old browsers or devices may not be fully supported.
Data Accuracy	The system depends on users and staff entering correct vehicle, mileage, booking, fuel, and payment information. Validation rules can reduce errors but cannot prevent all incorrect entries.
User Capacity	Expected peak user load and the exact number of service centres, fuel stations, and concurrent users must be finalized during detailed system planning.
Access Permissions	Detailed permissions for every staff role must be reviewed and approved before development to ensure appropriate access control.
Project Time	The system must be designed and developed within the academic semester; therefore, some advanced features will be postponed.
Future Features	Loyalty programmes, advanced analytics, group bookings, and a dedicated mobile application are outside the scope of Version 1.

Assumptions: The project assumes that users have access to a compatible device and internet connection, service centres maintain accurate operational information, and external providers offer available payment and messaging services.

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9. Project Timeline

Week	Planned Activity	Expected Output
Week 3	Finalize requirements, prepare proposal report, and conduct proposal presentation.	Approved proposal, requirements list, and project plan.
Week 4	Conduct detailed requirement analysis and prepare use-case diagrams.	Detailed requirements and initial use-case diagram.
Week 5	Design the database, prepare wireframes, and finalize the system architecture.	ER diagram/database design, UI wireframes, and architecture plan.
Week 6	Develop the User Management and Vehicle Management modules.	User registration, login, role management, and vehicle-record functions.
Week 7	Develop the Service Booking Management module.	Service booking, availability checking, cancellation, and rescheduling functions.
Week 8	Develop the Fuel Management module.	Fuel-recording, consumption-history, and discrepancy-detection functions.
Week 9	Develop the Payment Management and Notification Management modules.	Payment records, digital invoices, notifications, and reminders.
Week 10	Integrate completed modules, test key functions, and conduct the progress presentation.	Approximately 75% completed system and progress presentation.
Week 11	Complete remaining features and integrate external services where possible.	Integrated core system with remaining functions completed.
Week 12	Perform system testing, fix defects, improve the user interface, and prepare documentation.	Tested system, bug fixes, updated documentation, and final report draft.
Week 13	Conduct final presentation, system demonstration, and viva.	Fully functional system demonstration and final presentation.
Week 14	Submit the final report, source code, and supporting documentation.	Final report, source code, and project deliverables.

The team will use an Agile approach by dividing the work into manageable weekly tasks, reviewing progress regularly, integrating modules continuously, and resolving issues as early as possible.

10. Conclusion

AutoFlow Sri Lanka is proposed as a centralized web-based Vehicle Service and Fuel Management System that improves the way vehicle owners, service centres, fuel stations, and administrative staff manage daily operations. The system brings together vehicle records, service bookings, fuel records, payments, notifications, and user access management in one platform.

By replacing paper-based records, phone-call bookings, handwritten fuel logs, and disconnected payment processes, the proposed system will reduce manual work, minimize data-entry errors, prevent booking conflicts, and improve access to accurate information. Customers will benefit from easier service bookings, secure payments, digital invoices, and timely notifications. Staff and management will benefit from better coordination, controlled access, reliable records, and operational reports.

The proposed system is feasible to develop as a modular Java-based web application within the academic semester. The six major functions are assigned across the six group members, allowing the team to distribute responsibilities clearly and work in parallel. Through regular planning, development, testing, integration, and reviews, the team aims to deliver a practical and valuable solution for the vehicle-service and fuel-management domain.